

Air University



Organization & Assembly (Lab)

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Assignment # 01

Question 01:

Write a program to add two numbers and display the result in single digit, using a COM style program.

Code

```
ASM Question01.asm > ...
1  .MODEL TINY
2  .CODE
3      ORG 100h
4
5      START:
6          MOV AL, 05h    ; First number (5)
7          ADD AL, 03h    ; Add second number (3)
8          ADD AL, 30h    ; Convert to ASCII digit
9
10         MOV DL, AL     ; Prepare for output
11         MOV AH, 02h    ; Output character to STDOUT
12         INT 21h        ; Interrupt to DOS
13
14         MOV AH, 4Ch    ; Terminate program
15         INT 21h        ; Exit to DOS
16
17     END START
```

This program is designed to perform a simple addition and display the single-digit result on the screen.

- **Calculation:** It loads the number **5** into the **AL** register and adds **3** to it, making the result **AL = 8**.
- **Conversion:** It adds **30h** to the result, converting the numeric value **8** into its **ASCII** character equivalent, **'8'**.
- **Output:** It prepares the **DL** and **AH** registers and uses DOS Interrupt **21h**, Function **02h** to output the character **'8'** to the console.
- **Termination:** It uses DOS Interrupt **21h**, Function **4Ch** to cleanly exit the program.

Output

```
Z:\>mount d ./code
Drive D is mounted as local directory ./code/

Z:\>d:

D:\>set PATH=C:\TASM

D:\>TASM D:\test.asm
Turbo Assembler Version 4.1 Copyright (c) 1988, 1996 Borland International

Assembling file: D:\test.asm to test.OBJ
Error messages: None
Warning messages: None
Passes: 1
Remaining memory: 468k

D:\>TLINK D:\test
Turbo Link Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International
Warning: No stack

D:\>D:\test
8
D:\>I_
```

Question 02:

Write a program to subtract two numbers and display the result in single digit, using a COM style program.

Code

```
ASM Question02.asm > ...
1  .MODEL TINY
2  .CODE
3      ORG 100h
4
5  START:
6      MOV AL, 05h    ; First number (5)
7      SUB AL, 03h    ; Subtract second number (3)
8      ADD AL, 30h    ; Convert to ASCII digit
9
10     MOV DL, AL      ; Prepare for output
11     MOV AH, 02h     ; Output character to STDOUT
12     INT 21h         ; Interrupt to DOS
13
14     MOV AH, 4Ch     ; Terminate program
15     INT 21h         ; Exit to DOS
16
17     END START
```

This program is an assembly code snippet designed to calculate a simple subtraction and display the single-digit result.

- **Operation:** It performs the calculation $5 - 3$, storing the result (**AL = 2**) in the **AL** register.
- **Output Preparation:** It then adds **30h** to **AL**, converting the numeric result **2** into its ASCII character equivalent, **'2'**.
- **Display:** It prepares the **DL** and **AH** registers and uses DOS Interrupt **21h**, Function **02h** to display the character **'2'** on the screen.
- **Termination:** It uses DOS Interrupt **21h**, Function **4Ch** to terminate the program cleanly.

Output

```
Z:\>mount d ./code
Drive D is mounted as local directory ./code/

Z:\>d:

D:\>set PATH=C:\TASM

D:\>TASM D:\test.asm
Turbo Assembler Version 4.1 Copyright (c) 1988, 1996 Borland International

Assembling file: D:\test.asm to test.OBJ
Error messages:  None
Warning messages: None
Passes:         1
Remaining memory: 468k

D:\>TLINK D:\test
Turbo Link Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International
Warning: No stack

D:\>D:\test
2
D:\>|_
```

Question 03:

Write a program to multiply two numbers and display the result in single digit, using a COM style program.

Code

```
ASM Question03.asm > ...
1      .MODEL TINY
2      .CODE
3      |      |      |      ORG 100h
4      |
5      |      |      |      START:
6      |      |      |      MOV AL, 03h      ; First number (3)
7      |      |      |      MOV BL, 02h      ; Second number (2)
8      |      |      |      MUL BL           ; Multiply AL by BL
9      |      |      |      ADD AL, 30h      ; Convert to ASCII digit
10     |
11     |      |      |      MOV DL, AL       ; Prepare for output
12     |      |      |      MOV AH, 02h      ; Output character to STDOUT
13     |      |      |      INT 21h         ; Interrupt to DOS
14     |
15     |      |      |      MOV AH, 4Ch      ; Terminate program
16     |      |      |      INT 21h         ; Exit to DOS
17     |
18     |      |      |      END START
```

This assembly program is designed to perform a simple multiplication and display the single-digit result.

- **Multiplication:** It loads the factors **3** and **2** into registers **AL** and **BL** respectively. The instruction **MUL BL** multiplies **AL** by **BL** and stores the result (**AL=6**) back into **AL**.
- **Conversion and Output:** It adds **30h** to **AL**, converting the numeric result **6** into its **ASCII** character '**6**'. It then uses DOS Interrupt **21h**, Function **02h** to display the character in the **DL** register.
- **Termination:** The program uses DOS Interrupt **21h**, Function **4Ch** to terminate the program cleanly.

Output

```
Z:\>mount d ./code
Drive D is mounted as local directory ./code/

Z:\>d:

D:\>set PATH=C:\TASM

D:\>TASM D:\test.asm
Turbo Assembler Version 4.1 Copyright (c) 1988, 1996 Borland International

Assembling file:  D:\test.asm to test.OBJ
Error messages:   None
Warning messages: None
Passes:           1
Remaining memory: 468k

D:\>TLINK D:\test
Turbo Link Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International
Warning: No stack

D:\>D:\test
6 ←
D:\>I
```

THE END
